task-3

p-values

null hypothesis

① p-value < α ⇒ reject H_0

② $\geq \alpha$ ⇒ ~~accept~~ can't reject null hypothesis

prob of getting sample statistics – given H_0 true

assume

prob. sample.

significance level ⇒ $\alpha = 0.05$

⇒ Hypothesis testing

① Null hypothesis (H_0)

② Alternative (H_1)

no effect/ no difference (default assumption)

effect/diff

Goal : use data to decide whether there is enough evidence to doubt H_0

⇒ $\boxed{\text{P-value}}$

⇓

prob of observing your data (or something more extreme) if null hypothesis is true.

idea how surprising is my data?
assuming H_0 is true

↓ p-value → VERY SURPRISING ⇒ H_0 likely wrong

↑ p-value ⇒ not SURPRISING → H_0 still plausible

Ex H_0 ⇒ drug has no effect

Testing new drug → strong improvement

$p = 0.02$

↓

2%

↓

2% chance of seeing such results if drug actually had no effect

⇓

QUITE UNLIKELY
EVIDENCE AGAINST H_0

Significance test

⇓

converts the p value into a decision

① Choose a significance level (α)
predefined threshold
↳ 0.05 most common
↳ 0.01 (stricter)
↳ 0.001 (very strict)

② compare p value with α

$p \le \alpha$	reject H_0	evidence against null
$p > \alpha$	fail to reject H_0	not enough evidence

$\alpha = 0.05$
$p = 0.03$
$0.03 < 0.05$ (reject)

Result
→ $\boxed{\text{Statistically significant}}$

⇓

the observed result unlikely to be due to chance alone (under H_0)

α = threshold you choose beforehand
p = prob calc. from data.

⇒ p-value assumes H_0 already true & checks how likely the data is

⇒ $p < 0.005$, result imp XXX

⇓

only <u>stastically significant</u>

⇓ X practically imp.

<u>ex</u> A drug reduces blood pressure by 0.5 mmHg

$p = 0.001$ → stastically signif.

But effect tiny → not useful in real life

Stastistical ≠ practical imp

⇒ $p > 0.05$ means no effect exists XXX

⇓

means $\boxed{\text{insufficient evidence}}$

not proof of no effect

⇒ p-value : how shocking is my data if nothing is going on?

very shocking : reject H_0

not shocking : keep H_0